



Chapter 10

Image Segmentation

10.1 Fundamentals

- What is Image Segmentation?
- Background / Foreground

10.1 Fundamentals

- R: entire Image
- Partitions R into n sub regions R_1, R_2, R_n
 - (a) $\bigcup R_i = R$
 - (b) R_i is Connected set
 - (c) $R_i \cap R_j = \emptyset$ (*empty*)
 - (d) $Q(R_i) = \text{True}$ for $i=1, 2, \dots, n$
 - (e) $Q(R_i \cup R_j) = \text{false}$, for any adjacent regions R_i, R_j

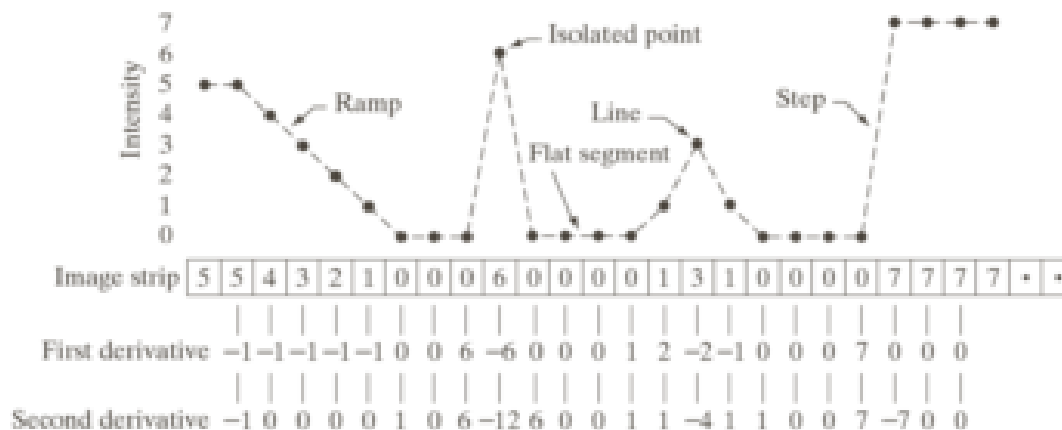
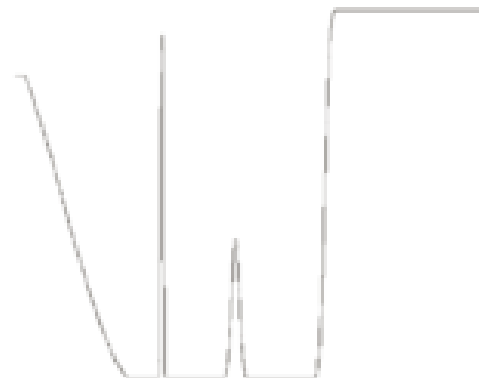
10.2 Point, Line and Edge Detection

- 10.2.1 Background

$$\frac{\partial f}{\partial x} = f(x+1, y) - f(x, y)$$

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

10.2.1 Background



10.2.2 Detection of Isolated Point

$$\begin{aligned}\nabla^2 f &= \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \\ &= f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)\end{aligned}$$

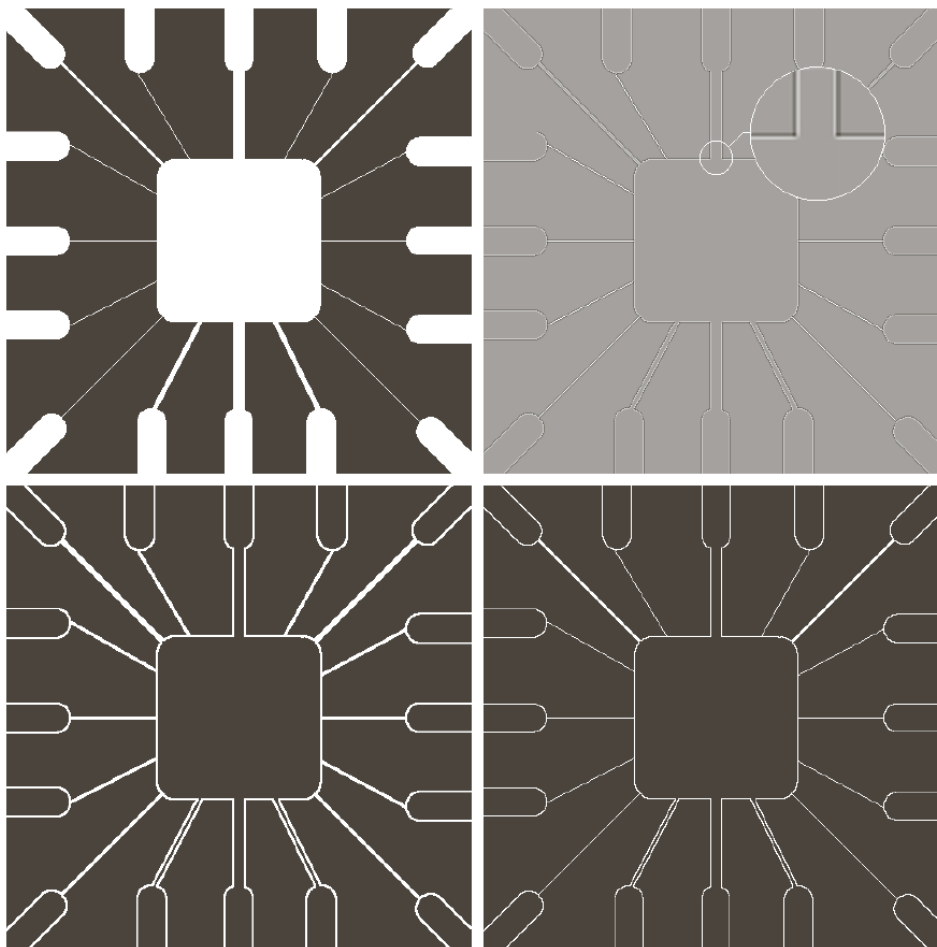
$$g(x, y) = \begin{cases} 1, & \text{if } |R(x, y)| > Thr \\ 0, & \text{otherwise} \end{cases}$$

1	1	1
1	-8	1
1	1	1



FIGURE 10.4
(a) Point detection (Laplacian) mask.
(b) X-ray image of turbine blade with a porosity. The porosity contains a single black pixel.
(c) Result of convolving the mask with the image. (d) Result of using Eq. (10.2-8) showing a single point (the point was enlarged to make it easier to see). (Original image courtesy of X-TEK Systems, Ltd.)

10.2.3 Line Detection(10.21)



a	b
c	d

FIGURE 10.5

(a) Original image.

(b) Laplacian image; the magnified section shows the positive/negative double-line effect characteristic of the Laplacian.

(c) Absolute value of the Laplacian.

(d) Positive values of the Laplacian.

10.2.3 Line Detection

- Line Detection Mask

-1	-1	-1	2	-1	-1	-1	2	-1	-1	-1	2
2	2	2	-1	2	-1	-1	2	-1	-1	2	-1
-1	-1	-1	-1	-1	2	-1	2	-1	2	-1	-1

Horizontal

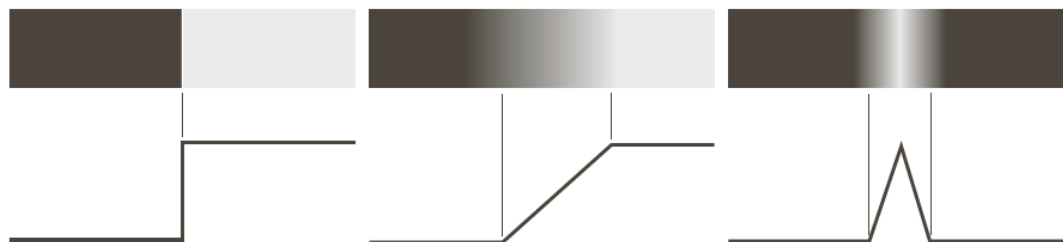
 $+45^\circ$

Vertical

 -45°

10.2.4 Edge Models

- Step, ramp and roof edge

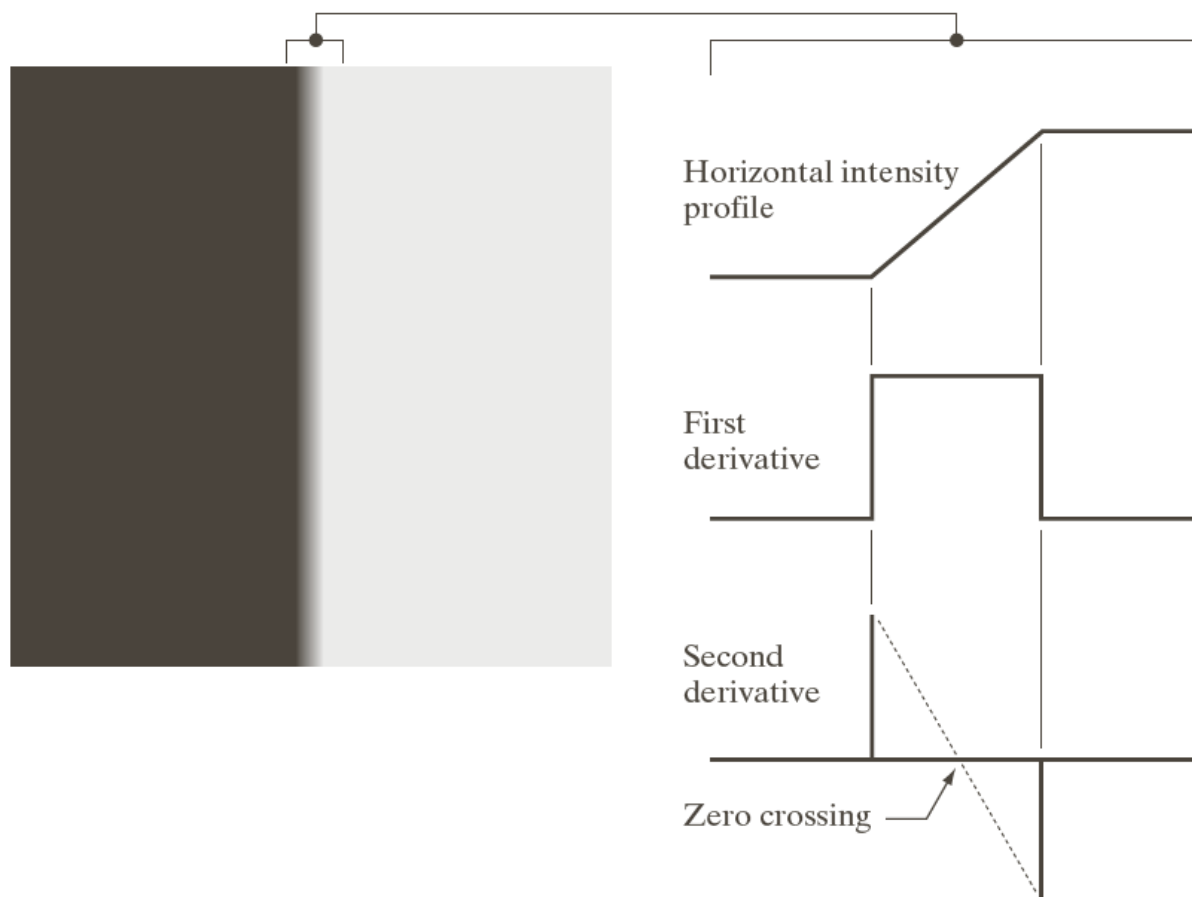


a b c

FIGURE 10.8

From left to right, models (ideal representations) of a step, a ramp, and a roof edge, and their corresponding intensity profiles.

10.2.4 Edge Models



a b

FIGURE 10.10

(a) Two regions of constant intensity separated by an ideal vertical ramp edge.
(b) Detail near the edge, showing a horizontal intensity profile, together with its first and second derivatives.

10.2.4 Edge Models

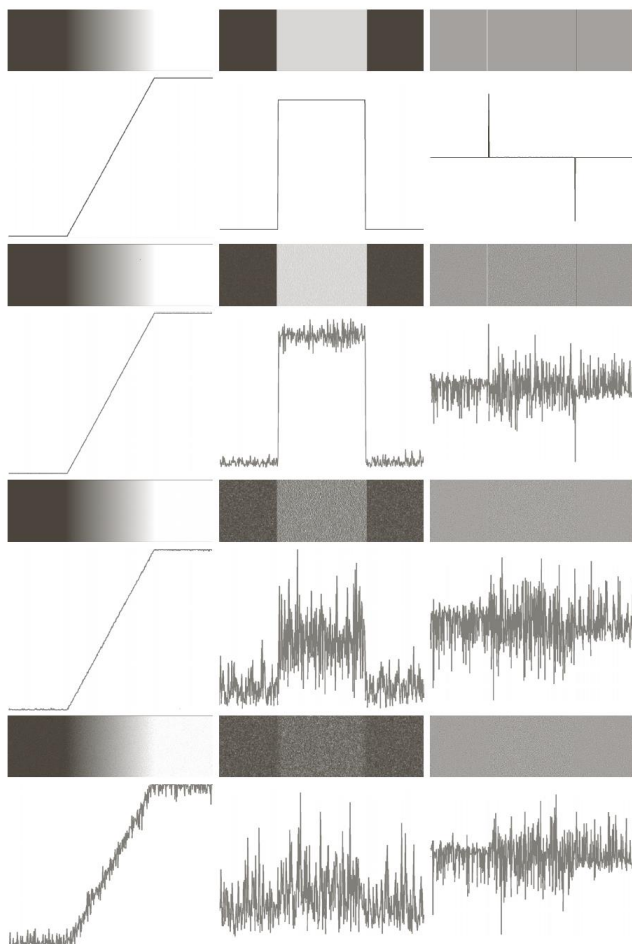


FIGURE 10.11 First column: Images and intensity profiles of a ramp edge corrupted by random Gaussian noise of zero mean and standard deviations of 0.0, 0.1, 1.0, and 10.0 intensity levels, respectively. Second column: First-derivative images and intensity profiles. Third column: Second-derivative images and intensity profiles.

10.2.5 Basic Edge Detection

- The image gradient and its properties
 - Gradient

$$\nabla f = \text{grad}(f) = \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

The image gradient and its properties

- Magnitude (length) of gradient vector

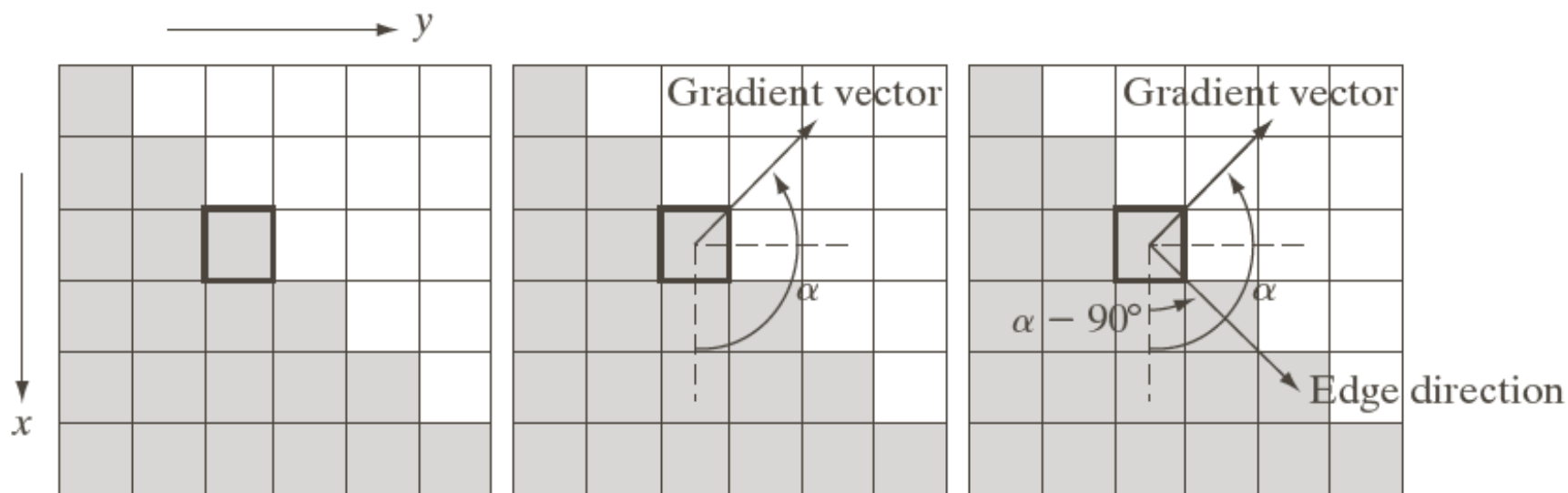
$$M(x, y) = \text{mag}(\nabla f) = \sqrt{g_x^2 + g_y^2}$$

- Direction of gradient vector

$$\alpha(x, y) = \tan^{-1} \left[\frac{g_y}{g_x} \right]$$

$$\varphi(x, y) = a \tan 2 (g_y(x, y), g_x(x, y))$$

The image gradient and its properties



a b c

FIGURE 10.12 Using the gradient to determine edge strength and direction at a point. Note that the edge is perpendicular to the direction of the gradient vector at the point where the gradient is computed. Each square in the figure represents one pixel.

Gradient operators

$$g_x = \frac{\partial f(x, y)}{\partial x} = f(x + 1, y) - f(x, y)$$

$$g_y = \frac{\partial f(x, y)}{\partial y} = f(x, y + 1) - f(x, y)$$

-1
1

-1	1
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a b

FIGURE 10.13
One-dimensional
masks used to
implement Eqs.
(10.2-12) and
(10.2-13).

Gradient operators

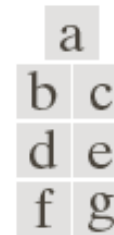
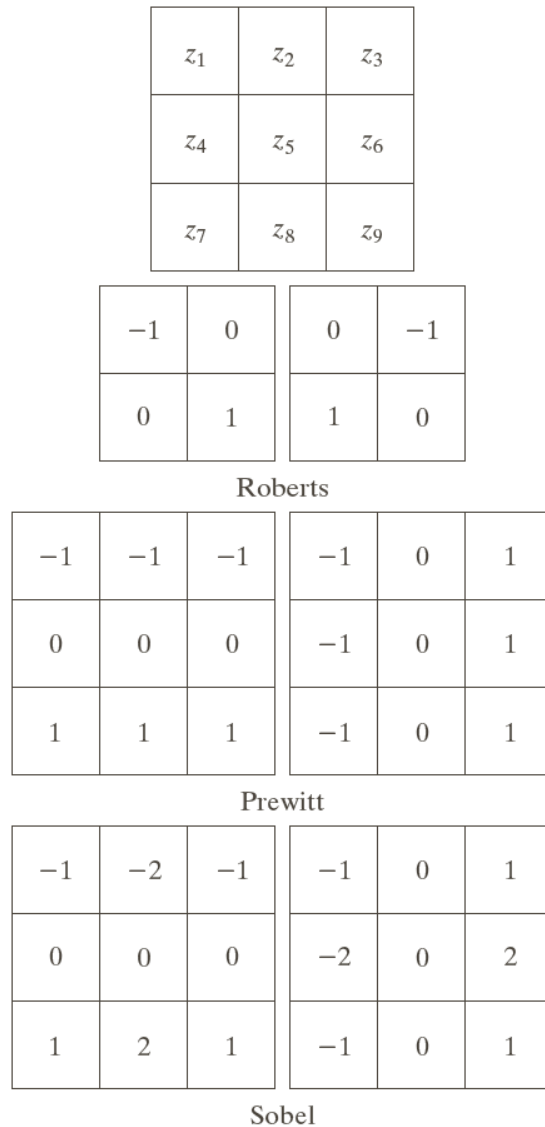


FIGURE 10.14
A 3×3 region of an image (the z 's are intensity values) and various masks used to compute the gradient at the point labeled z_5 .



a	b
c	d

FIGURE 10.16

(a) Original image
of size

834×1114 pixels,
with intensity
values scaled to
the range $[0, 1]$.

(b) $|g_x|$, the
component of the
gradient in the
 x -direction,
obtained using
the Sobel mask in
Fig. 10.14(f) to
filter the image.

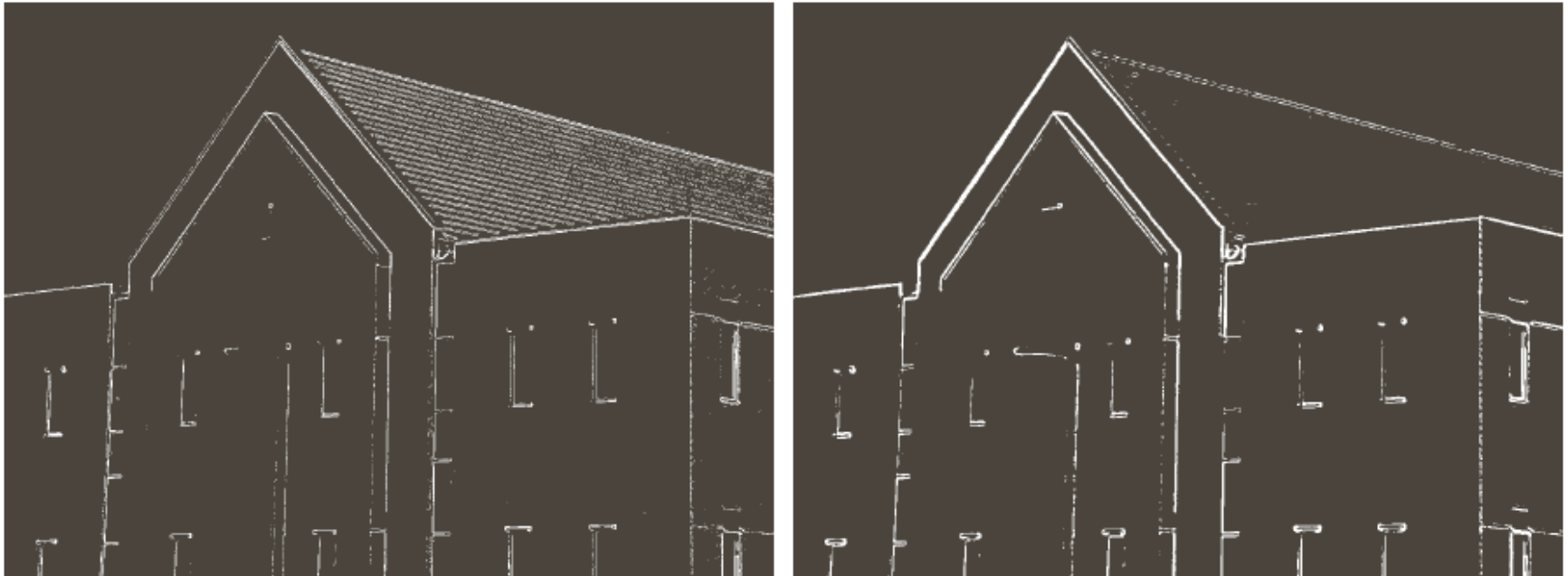
(c) $|g_y|$, obtained
using the mask in
Fig. 10.14(g).

(d) The gradient
image, $|g_x| + |g_y|$.

Combining the gradient with thresholding

$$e(x, y) = \begin{cases} 1, & M(x, y) > \text{Thr} \\ 0, & \textit{otherwise} \end{cases}$$

Combining the gradient with thresholding



a b

FIGURE 10.20 (a) Thresholded version of the image in Fig. 10.16(d), with the threshold selected as 33% of the highest value in the image; this threshold was just high enough to eliminate most of the brick edges in the gradient image. (b) Thresholded version of the image in Fig. 10.18(d), obtained using a threshold equal to 33% of the highest value in that image.